SYSTEM TEST REPORT

Hospital Management System

1. Introduction

System testing evaluates the behaviour of the Hospital Management System as a complete application, focusing on user workflows and the interaction between frontend screens and backend services. This report is prepared from the supplied project ZIP, its application source, and the UI screenshots included in the repository.

The project includes patient registration and OTP verification, authentication, doctor availability and appointment booking, appointment cancellation, bed availability, patient bills and online payment, medical history, diagnosis reports, prescription management, and medical file download. The supplied backend is implemented with Node.js and Express; the frontend uses React and Vite.

2. Objectives of System Testing

• Verify the main user-facing workflows represented in the Hospital Management System.

• Check the expected interaction between frontend pages and backend API endpoints.

• Review authentication and OTP verification flows.

• Check appointment, bed, billing, medical-record, prescription, and file-download workflows.

• Identify evidence gaps and distinguish documented UI evidence from tests that still require execution.

3. Testing Environment

Component	Project details
Application	Hospital Management System
Frontend	React 19, Vite 8
Backend	Node.js with Express 5
Authentication / Security Libraries	bcryptjs, jsonwebtoken, OTP verification
Other Backend Libraries	cors, dotenv, multer
Frontend Test Framework in supplied ZIP	No frontend test script or test framework is configured in the supplied ZIP's frontend package.json.
Backend Test Framework in supplied ZIP	No backend test suite is configured; backend package.json contains a placeholder test script.
Data Storage	Backend source uses in-memory arrays for application data; no persistent database configuration is evidenced in the supplied server source.
Execution Environment	Not independently executed while preparing this report; intended local environment is Node.js/npm with browser-based frontend testing.

Environment note: the supplied ZIP does not contain the Vitest test files or Vitest configuration discussed in the earlier frontend testing work. The 15 passing frontend unit tests previously reported belong to that separate working copy and are not counted as system-test executions in this report.

4. System Test Scenarios and Test Cases

The following scenarios are derived from the application modules and screenshot evidence present in the supplied ZIP. They define the system-level workflows to verify. A screenshot in the repository demonstrates a UI state, but by itself does not prove that the complete workflow or backend operation passed.

Test ID	Scenario	Test procedure	Expected result	Evidence / status
ST-001	Patient registration	Submit valid patient details and register an account.	Registration response and next verification step are shown.	Patient registration screenshot supplied; execution status not independently verified.
ST-002	OTP verification	Enter the OTP associated with a registered phone number.	Phone number is marked verified or a validation error is shown.	OTP verification screenshot supplied; execution status not independently verified.
ST-003	Doctor availability and booking	Select a doctor, date, and available appointment slot; submit booking.	Available slots are shown and a valid booking is confirmed.	Doctor availability / appointment booking screenshot supplied; execution status not independently verified.
ST-004	Appointment cancellation	Cancel an eligible booked appointment.	Appointment status changes to cancelled and the updated list is shown.	Appointment cancellation screenshot supplied; execution status not independently verified.
ST-005	Upcoming and past appointments	Open the appointments view.	Upcoming and past appointment records are displayed in the appropriate sections.	Appointments screenshot supplied; execution status not independently verified.

Test ID	Scenario	Test procedure	Expected result	Evidence / status
ST-006	Bed availability	Open the bed availability view.	Bed status and availability information are displayed.	Bed availability screenshot supplied; execution status not independently verified.
ST-007	Patient bills	Open the patient's billing page.	The patient's bill information is displayed.	Patient bills screenshot supplied; execution status not independently verified.
ST-008	Online payment	Initiate payment for an eligible bill.	Payment workflow responds with an appropriate result and updated payment state.	Online payment screenshot supplied; execution status not independently verified.
ST-009	Patient medical history	Open the patient's medical history.	Relevant medical history records are displayed to the authorized patient.	Medical history screenshot supplied; execution status not independently verified.
ST-010	Medical diagnosis reports	Open diagnosis/report information.	Diagnosis and report details are displayed to the authorized user.	Diagnosis reports screenshot supplied; execution status not independently verified.
ST-011	Prescription management	Open prescription information.	Prescription details and associated medicine information are displayed.	Prescription screenshot supplied; execution status not independently verified.
ST-012	Medical file download	Request a medical file/report download.	The application returns a downloadable file for the selected record.	Medical file download screenshot supplied; execution status not independently verified.

5. Test Execution Summary

The supplied project ZIP contains application source code and twelve named UI screenshots. However, it does not contain a recorded system-test execution log, a system-test automation suite, or a verified pass/fail summary for the workflows listed in Section 4.

Parameter	Result
System test scenarios documented	12
UI screenshots supplied in repository	12
System test executions verified for this report	Not established from supplied ZIP
Passed / Failed / Blocked	Not reported — requires execution evidence
Overall system test result	Pending execution and evidence review

This is intentional: the report does not convert screenshots or previously reported frontend unit-test results into unverified system-test passes.

6. Testing Evidence

The following application screenshots were included in the supplied project ZIP and are inserted below as UI evidence. They should be supplemented with fresh screenshots of the application running against the synchronized project version, plus terminal/API evidence for the actual system-test execution.

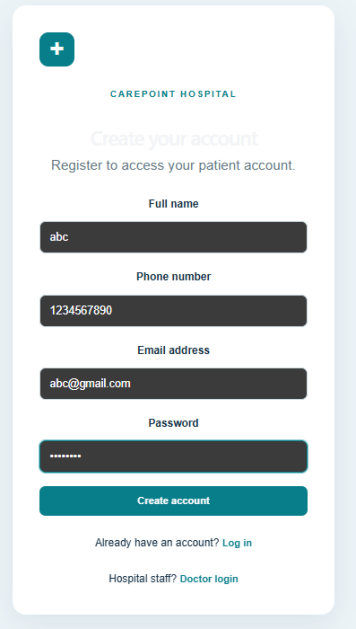


Figure 1. Patient registration screen.

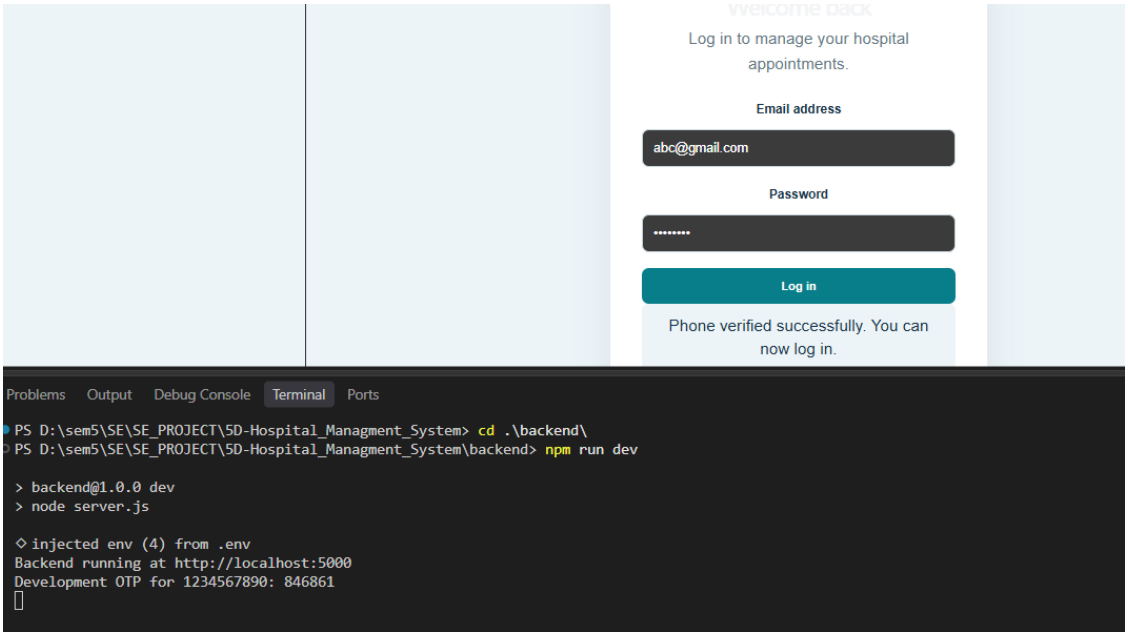


Figure 2. OTP verification screen.

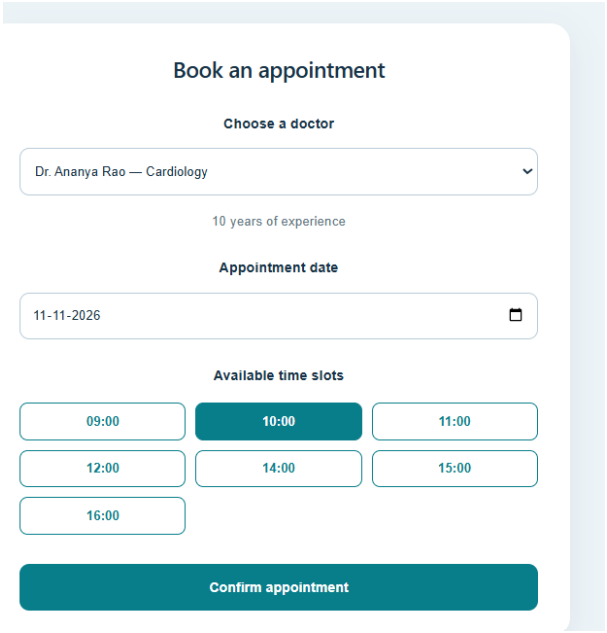


Figure 3. Doctor availability and appointment booking screen.

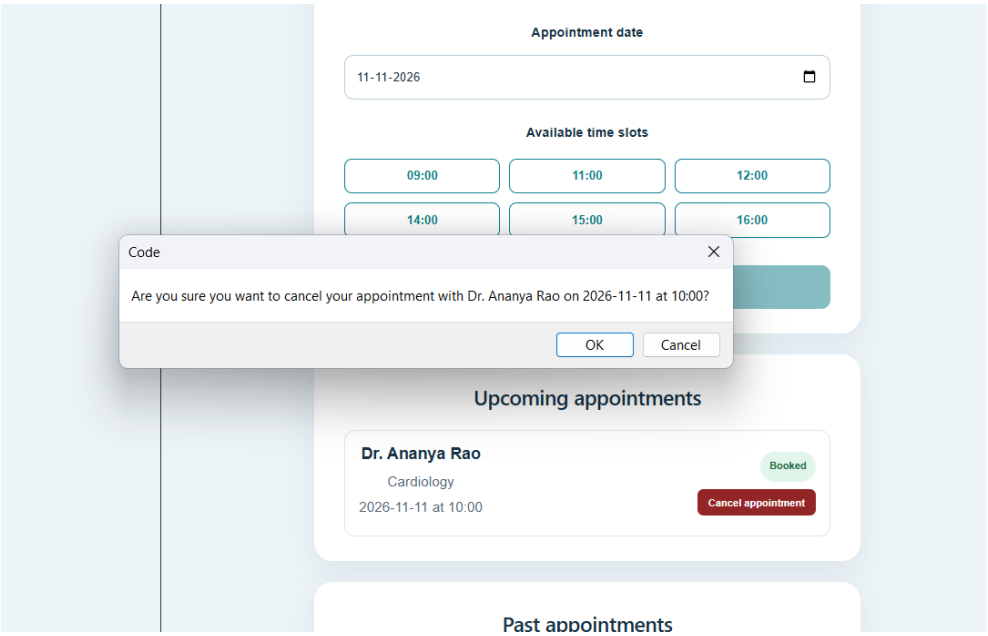


Figure 4. Appointment cancellation screen.

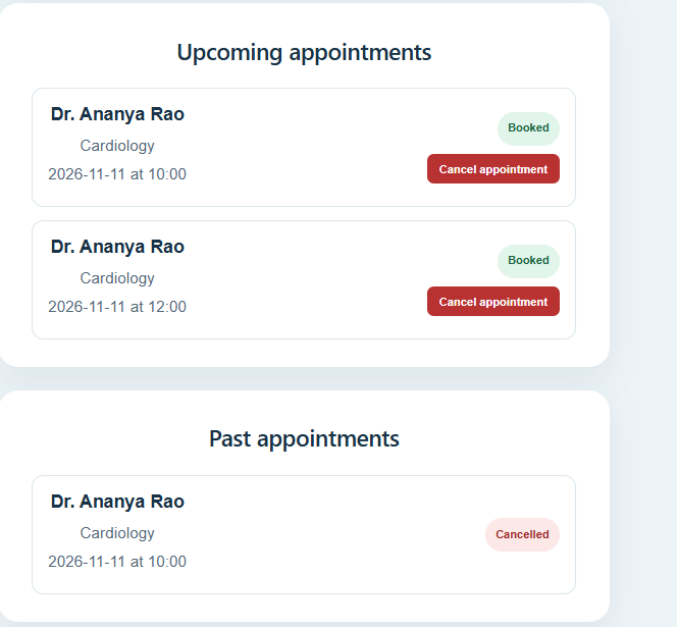


Figure 5. Upcoming and past appointments screen.

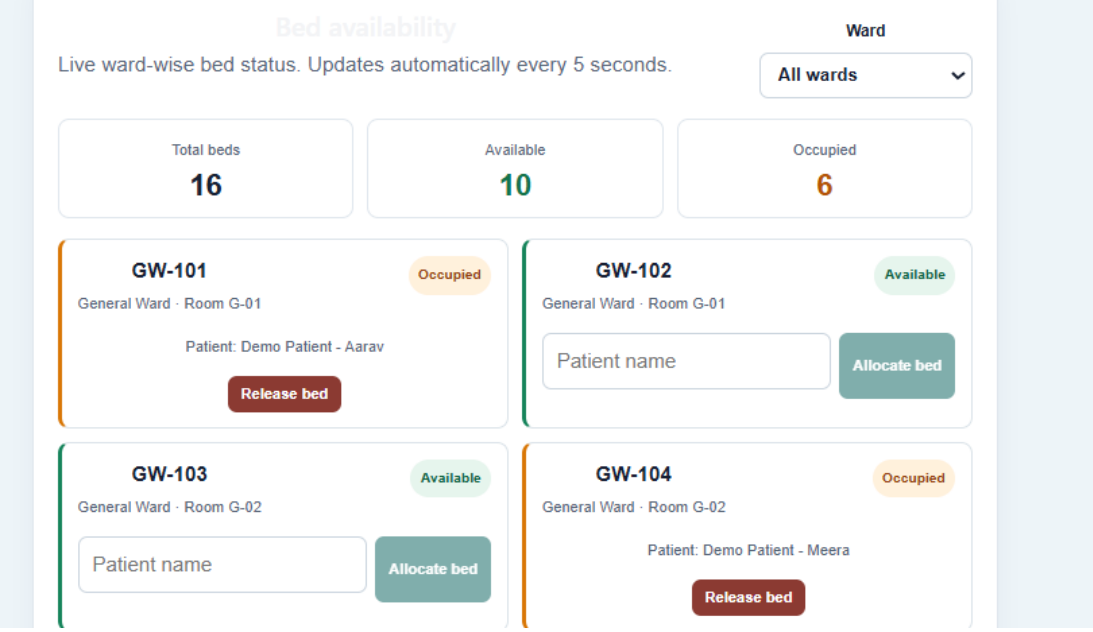


Figure 6. Bed availability screen.

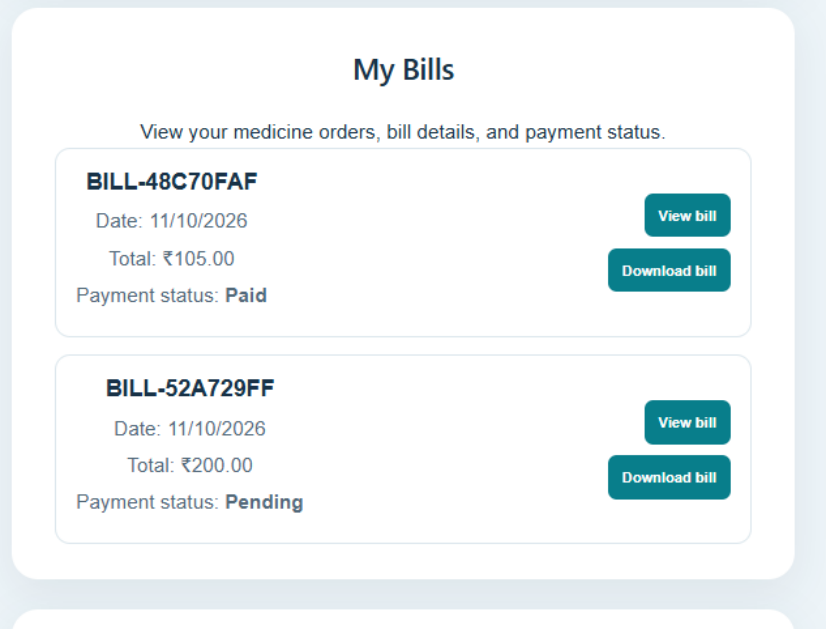


Figure 7. Patient bills screen.

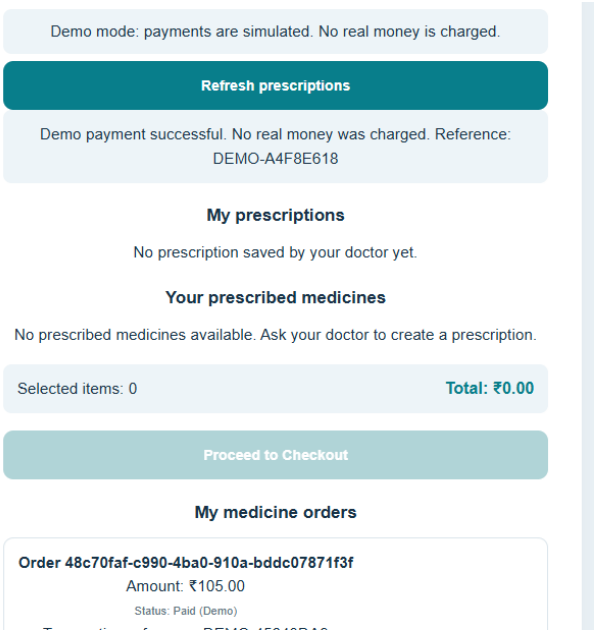


Figure 8. Online payment screen.

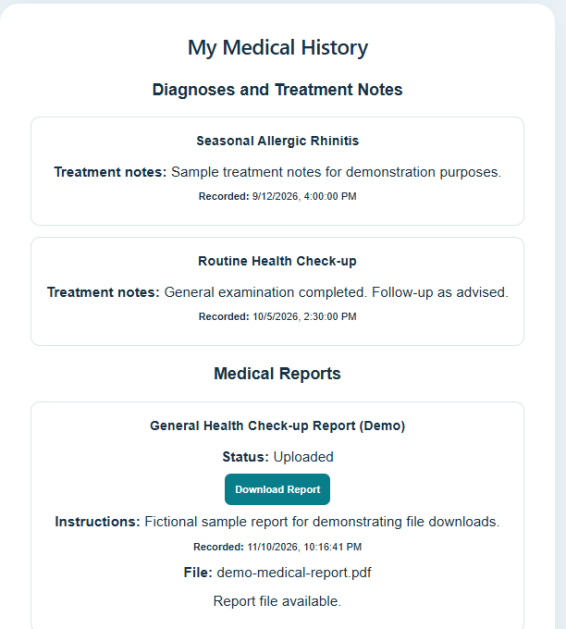


Figure 9. Patient medical history screen.

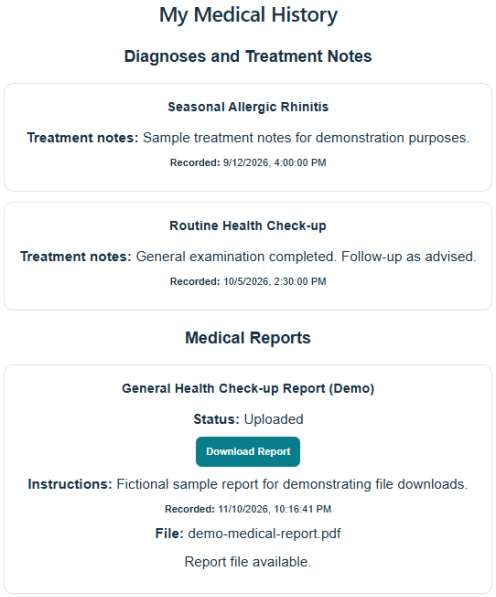


Figure 10. Medical diagnosis reports screen.

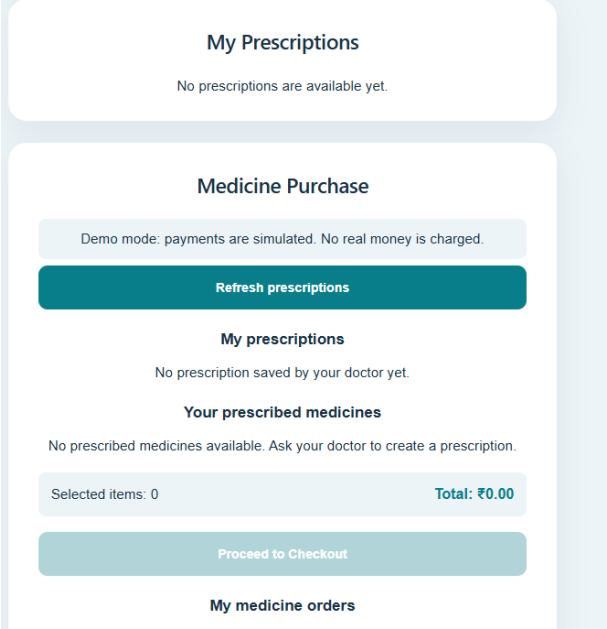


Figure 11. Prescription management screen.

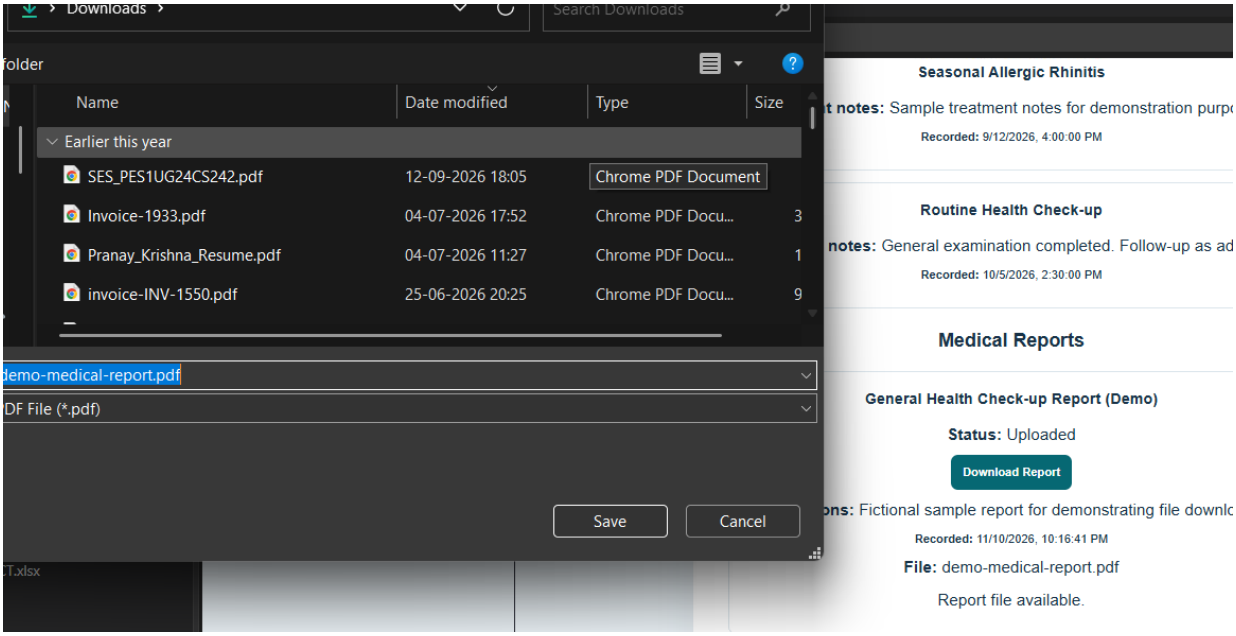


Figure 12. Medical file download screen.

7. Conclusion

The supplied Hospital Management System project contains frontend and backend implementation for patient registration, OTP verification, appointment scheduling and cancellation, bed availability, billing and payment, medical history, diagnosis reports, prescriptions, and medical file downloads. The repository also includes twelve screenshots illustrating these application areas.

The supplied archive does not provide a verified system-test run or system-level pass/fail results. Therefore, this report documents the identified system scenarios and available UI evidence, while leaving the execution outcome pending. To finalize system testing, run the application from the synchronized repository, execute each applicable end-to-end workflow, and capture the resulting browser and terminal/API evidence. Only then should individual scenarios be marked PASS or FAIL.

Source basis: supplied 5D-Hospital_Managment_System-main.zip, including backend/server.js, frontend/package.json, backend/package.json, and the repository screenshots folder.